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**CHARACTERISTICS OF HUMAN INFLUENZA AH1N1, AH3N2 AND B
VIRUSES ISOLATED FEBRUARY TO AUGUST 2007**

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Influenza Activity worldwide, February to September 2007

During February to September 2007, influenza AH1N1, AH3N2 and B viruses continued to circulate worldwide and cause outbreaks in many countries. Influenza AH3N2 viruses predominated overall and comprised roughly half of the viruses isolated; in some countries, such as Argentina (during June) and Australia (during August) influenza activity due to H3N2 viruses was higher than in recent years. AH1N1 viruses caused outbreaks in the United States, where they predominated early in 2007, and more recently in, for example, New Zealand and South Africa. Outbreaks due to influenza B were reported in, for example, Brazil, Chile and Japan and the majority of viruses isolated in Madagascar were influenza B. Whereas B/Victoria/2/87 lineage viruses predominated in many countries, the proportion of B/Yamagata/16/88 lineage viruses increased and recently they were predominant among B viruses isolated in Australia, Chile, Hong Kong SAR China, New Zealand and some Asian countries.

Antigenic and genetic characteristics of influenza viruses, isolated February to August 2007

Approximately 500 influenza viruses, isolated in 29 countries during February to August 2007, were characterized at the WHO Influenza Centre. Roughly half (49%) were H3N2, 24% were H1N1 and 27% were influenza B (Table 1).

AH1N1 viruses

The H1N1 viruses were antigenically closely related to either A/New Caledonia/20/99 (A/Thessaloniki/24/2005 or A/St.Petersburg/10/2007) or A/Solomon Islands/3/2006 (A/Hong Kong/2562/2006 or A/Fukushima/141/2006 (Tables 2 and 3), the relative proportions of which varied in different countries. The A/New Caledonia/20/99-like viruses accounted for 61% of isolates received and were predominant among viruses isolated in e.g. Georgia, Russia, Spain, Ukraine and the UK, whereas A/Solomon Islands/3/2006-like viruses (39%) predominated in e.g. Italy, Mauritius, Norway, Poland and South Africa. Of the viruses isolated during February, which comprised roughly half of the total, a third were A/Solomon Islands/3/2006-like; the proportion of A/Solomon Islands/3/2006-like viruses received increased to approximately half of the viruses isolated during March to August. Recently A/Solomon Islands/3/2006-like viruses have predominated in east Asia, Australia and New Zealand.

As previously, the **HA sequences** fell into two principal clades (Figure 1).

Clade 1, represented by A/Thessaloniki/24/2005, was characterized by the amino acid change Y252F, relative to the HA sequence of A/New Caledonia/20/99. There has been relatively little change in these HA sequences

and the corresponding NA sequences of viruses isolated during the past two years (Figures 1 and 2; Table 4).

HA sequences within **Clade 2**, which are characterized by the 5 common amino acid changes T82K, Y94H, R145K, R208K and T266N, exhibit greater diversity and include 3 principal sublineages, represented by: A, A/Solomon Islands/3/2006 (and A/Fukushima/141/2006); B, A/St Petersburg/96/2007; C, A/Hong Kong/2652/2006 (Figure 1). Despite differences in HA sequence, there was a close correlation between the antigenicity of A/Solomon Islands/3/2006-like viruses (including A/HK/2652/2006 and A/Fukushima/141/2006) and the presence of the amino acid change K140E, irrespective of the clade/sublineage into which the HA fell. Although most of the HA sequences within sublineages A and B, of viruses isolated during 2007, possess E140, a significant subgroup within sublineage C retain K140 and are antigenically closely related to A/New Caledonia/20/99 and A/St Petersburg/10/2007 (Table 3).

Equivalent heterogeneity to that in HA was observed in the divergence of corresponding sublineages of **NA sequences** (Figure 2). Divergence in NA sequence was greater than that of HA sequence; for example, there were 16 common amino acid differences between the NAs of A/Solomon Islands/3/2006 and A/Hong Kong/2652/2006 (Figure 2; Table 4).

M2 protein gene sequences obtained for the majority of selected viruses with HAs in clades 1 and 2 did not contain amantadine resistance mutations. As previously, viruses with HAs which fell into the sublineage C, represented by A/HK/2652/2006, possessed the resistance mutation S31N (as indicated on Figure 1), indicative of resistance to amantadine and rimantadine.

AH3N2 viruses

Although some of the H3N2 viruses were shown to be antigenically closely related to the A/Wisconsin/67/2005-like reference viruses, many gave low HI titres, relative to the homologous virus, with antisera to the egg-grown vaccine strains, A/Wisconsin/67/2005 and A/Hiroshima/52/2005, and more recent reference viruses (Tables 5-8). MDCK cell-grown reference viruses, such as A/Parma/1/2007 and A/Trieste/25c/2007, also gave relatively low homologous HI titres with their corresponding ferret antisera, compared with egg-passaged reference viruses. However, ferret antisera to these MDCK cell-grown viruses gave similar patterns of reactivity with the panel of reference viruses as antisera to egg-grown viruses (compare eg. antisera to A/Trieste/25c/2007 and the egg passaged variant A/Trieste/25E/2007, Tables 7 and 8), and did not readily distinguish between egg-passaged recent isolates, with HAs which fell within the principal clades. Titres were similar in HI tests using guinea pig or human red blood cells in place of turkey RBCs.

The majority of **HA sequences** fell within one of three principal clades, distinguished by the amino acid changes relative to A/Wisconsin/67/05: A, G50E and K140I, represented by A/Brisbane/10/2007 (A/Trieste/25/2007 or A/Wisconsin/3/2007); B, N6I, R142G, L157S and K173E, represented by A/Nepal/921/2006 (A/Parma/1/2007, A/Serres/77/2007 or A/Prague/3/2007); and C, R142G and N144D, represented by A/Denmark/4/2007 (or A/Henan/Jinshui/147/2007) (Figure 3).

In general **NA sequences** of these viruses were shown to fall into corresponding clades (Figure 4, Table 9).

M2 sequences of selected viruses showed that, in contrast to earlier “Wisconsin” viruses, the majority of which possessed the amantadine resistance mutation S31N, approximately half of the more recent isolates studied lacked known resistance mutations. The “Nepal” clade (B) were mainly sensitive, whereas clade A and C viruses analysed retained the S31N resistance marker in M2 (Figure 3).

B viruses

The majority of B viruses received were of the B/Victoria lineage; however, the relative proportions of B/Victoria and B/Yamagata lineage viruses isolated in different countries varied. For example, B/Victoria lineage viruses predominated in Madagascar, Russia, South Africa and the Ukraine, whereas B/Yamagata lineage viruses were more prevalent among isolates in Albania, Belarus and Norway. More recently, B/Yamagata lineage viruses have predominated in Australia, Chile, Hong Kong SAR China and New Zealand.

The **B/Victoria lineage viruses** were antigenically closely related to B/Hong Kong/45/05, the vaccine strain B/Malaysia/2506/04 and the more recent reference strain B/Victoria/304/06, although HI titres for many of the isolates were reduced with antisera to the latter two egg-grown viruses (Table 10).

Their **HA sequences** were close to those of B/Victoria lineage viruses isolated during 2006 and B/Malaysia/2506/2004 (Figure 5). Most of the **NA sequences** obtained for viruses isolated during 2006 and 2007 differed from that of B/Malaysia/2506/2004 by 8 amino acids (Figure 7; Table 12).

The **B/Yamagata lineage viruses** were, in general, antigenically closely related to the reference viruses B/Florida/7/2004, B/Egypt/144/2005 and B/Florida/4/2006 (Table 11).

HA sequences of recent B/Yamagata lineage isolates were similar to those of viruses isolated during 2006 and showed only minor variation (Figure 6). Amino acid differences in HA and NA, characteristic of sequence variants, are listed in Table 12.

Table 1. Human influenza isolates, February 2007 - present

Country	H1N1		H3N2	B	
	A/NC 20/99	A/SI 3/06		Yamagata lineage	Victoria lineage
February					
Albania				4	
Austria			6		
Belarus	1		3	1	
Bulgaria			9		
Croatia	2		2		
Czech Republic	1		17		
France	2	1	4		
Georgia	4		7		1
Germany	3		17		
Iceland	3	2	14		
Italy	1	9	11	2	2
Latvia	1		2		
Madagascar					33
Norway		3	5	1	
Poland		4			
Portugal			18	1	
Russia	3			1	5
South Korea				1	
Spain	3		3		
Sweden			1		
Switzerland			2		
Ukraine	14		16		16
United Kingdom	5	1	8		
March					
Albania				5	2
Belarus			11	2	
France	2				
Georgia	3		8		1
Hong Kong	1	1	1		1
Iceland	1	1	6		
Latvia	1		3		
Madagascar				1	25
Norway		2	5	2	
Poland		4	1		
Ukraine	1		11		1
Russia	4	1	4		1
April					
Georgia			4		
Hong Kong		1		1	
Iceland		1	1		
Latvia			1		
Mauritius			1		
Russia	9	3	15		4
Sweden			11	1	
May					
Hong Kong			1	1	
Iceland		1			
Mauritius			1		
Russia	3	2	6		
Senegal					6
June					
Hong Kong		1			
Mauritius					1
Russia	3			1	1
South Africa					2
July					
Mauritius		3		1	2
South Africa	1	4	6		5
United Kingdom		1			
August					
Mauritius		1			
Total = 496	72	47	242	26	109
	119 (24%)		(49%)	135 (27%)	

Table 2. Antigenic analyses of influenza A H1N1 viruses

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre ¹						
			Post infecton ferret sera						
			A/NC 20/99	A/Thess 24/05	A/Egypt 39/05	A/Fuk 97/06	A/HK 2652/06	A/SI 3/06	A/Fuk 141/06
A/New Caledonia/20/99	9.6.99	Ex	320	640	640	160	80	40	40
A/Thessaloniki/24/05	3.2.05	E2 \3	320	1280	1280	320	160	80	80
A/Egypt/39/2005	25.11.05	E1 \4	80	320	1280	80	40	80	80
A/Fukushima/97/2006	23.1.06	MDCK4 \1	40	40	160	640	640	640	640
A/Hong Kong/2652/2006	17.7.06	MDCK2 \1	<	<	80	320	640	320	320
A/Solomon Islands/3/2006	21.8.06	E3 \2	40	160	160	160	320	640	320
A/Fukushima/141/2006	7.2.06	E4 \2	40	40	320	640	320	640	320
A/Georgia/16/2007	21.2.07	MDCK2 \1	80	320	320	40	80	<	40
A/Cherkasy SES/136/07	26.2.07	MDCKx \1	160	640	640	80	80	160	160
A/Paris/1738/2007	Mar-07	MDCKx \1	160	1280	640	80	80	40	80
A/Iceland/43/07	9.3.07	MDCK2 \1	160	640	640	80	40	<	40
A/Hong Kong/1759/2007	21.3.07	MDCK2 \1	160	640	640	40	160	40	40
A/Latvia/4318/07	21.3.07	MDCK1 \1	160	1280	1280	80	40	40	160
A/St.Petersburg/83/07	9.4.07	C1 \2	160	640	1280	80	80	40	80
A/Tula/5/07	9.4.07	C1 \1	160	1280	640	80	40	<	80
A/Astrakhan/80/2007	21.5.07	C2 \2	160	640	640	40	160	40	40
A/Cape Town/106/07	10.7.07	MDCKx \1	160	1280	640	160	80	40	80
A/Ulaanbaatar/2212/2007	unknown	C3 \1	<	<	40	160	320	160	160
A/Vienna/337855/07	5.2.07	MDCK1 \1	<	<	80	1280	80	640	320
A/Incheon/2647/07	7.2.07	MDCK2 \1	<	<	40	80	160	160	160
A/Norway/651/2007	8.3.07	LLC-MK2, MDCK1 \1	40	80	160	320	160	320	320
A/Warsaw/9/07	9.3.07	E \1	40	80	320	320	640	640	320
A/St.Petersburg/24/07	12.3.07	E2 \1	<	<	<	160	160	160	160
A/Khabarovsk/499/07	2.4.07	C1 \1	<	40	160	80	320	80	160
A/Malaysia/472/2007	4.4.07	MDCKx+1 \1	<	<	<	640	320	320	160
A/Singapore/26/2007	11.4.07	MDCK2 \1	<	80	40	160	320	160	160
A/Iceland/49/07	13.4.07	MDCK2 \1	<	<	80	80	160	320	160
A/Hong Kong/2387/2007	30.4.07	MDCK2 \1	<	80	40	80	320	160	80
A/St Petersburg/96/2007	21.5.07	C2 \1	<	<	80	160	320	160	320
A/Hong Kong/2613/2007	5.6.07	MDCK2 \1	<	80	40	80	320	160	160
A/Johannesburg/80/07	10.7.07	MDCKx \1	<	80	40	320	640	320	80
A/Johannesburg/141/07	21.7.07	MDCKx \1	<	80	40	160	640	320	160
A/Mauritius/543/2007	25.7.07	MDCK3 \1	<	40	80	160	160	160	320

1. < : <40

Table 3. Antigenic analyses of influenza A H1N1 viruses

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre ¹							
			Post infecton ferret sera							
			A/NC 20/99	A/Thess 24/05	A/Egypt 39/05	A/S. Peter 10/07	A/Fuk 97/06	A/HK 2652/06	A/SI 3/06	A/Fuk 141/06
A/New Caledonia/20/99	9.6.99	Ex	320	640	640	160	160	80	40	80
A/Thessaloniki/24/05	3.2.05	E2 \3	320	1280	1280	640	320	80	80	160
A/Egypt/39/2005	25.11.05	E1 \4	160	640	640	160	160	<	40	160
A/St. Petersburg/10/2007	16.1.07	MDCK1 \2	80	320	320	640	40	80	<	<
A/Fukushima/97/2006	23.1.06	MDCK4 \1	40	40	80	<	320	640	320	640
A/Hong Kong/2652/2006	17.7.06	MDCK2 \1	40	160	80	<	640	320	640	640
A/Solomon Islands/3/2006	21.8.06	E3 \2	40	80	320	<	640	320	640	640
A/Fukushima/141/2006	7.2.06	E4 \2	40	80	320	<	640	320	640	640
A/Pennsylvania/1/2007	unknown	E1 \1	160	1280	640	320	40	40	40	<
A/St.Petersburg/43/07-E	19.3.07	E2 \1	80	320	160	320	40	80	40	40
A/Cape Town/106/07	10.07.07	MDCK \1	160	1280	640	320	160	80	40	80
A/Ulaanbaatar/116/2007	unknown	E3 \1	<	<	80	40	320	320	320	320
A/Texas/5/2007	unknown	M1,C1 \1	40	40	80	<	160	160	320	320
A/Kitakyusyu/6/2007	unknown	MDCK4 \1	<	<	40	<	80	160	80	160
A/Peru/1169/2007	unknown	SpfCk1,E2 \1	<	40	320	<	320	160	320	320
A/Malaysia/472/2007	4.4.07	MDCKx+1 \1	<	<	<	<	640	320	320	160
A/St Petersburg/96/2007	21.5.07	C2 \1	40	160	160	40	320	320	320	320
A/Sydney/207/2007	26.6.07	MDCK3 \1	<	80	40	<	160	160	160	80
A/Johannesburg/67/07	9.7.07	MDCK \1	<	<	40	<	640	320	320	320
A/Mauritius/507/2007	17.7.07	MDCK3 \1	<	80	<	<	160	80	80	80
A/Johannesburg/148/07	23.7.07	MDCK \1	<	160	40	<	160	160	160	160
A/Mauritius/610/2007	13.8.07	MDCK2 \1	<	40	40	<	80	80	80	160

1. < : <40

Figure 1. Phylogenetic comparison of H1 HA genes

Vaccine strain

Reference strains

Isolates

July-December 2006

January-March 2007

April-July 2007

* Amantadine resistant
(S31N in M2)

● Amantadine sensitive

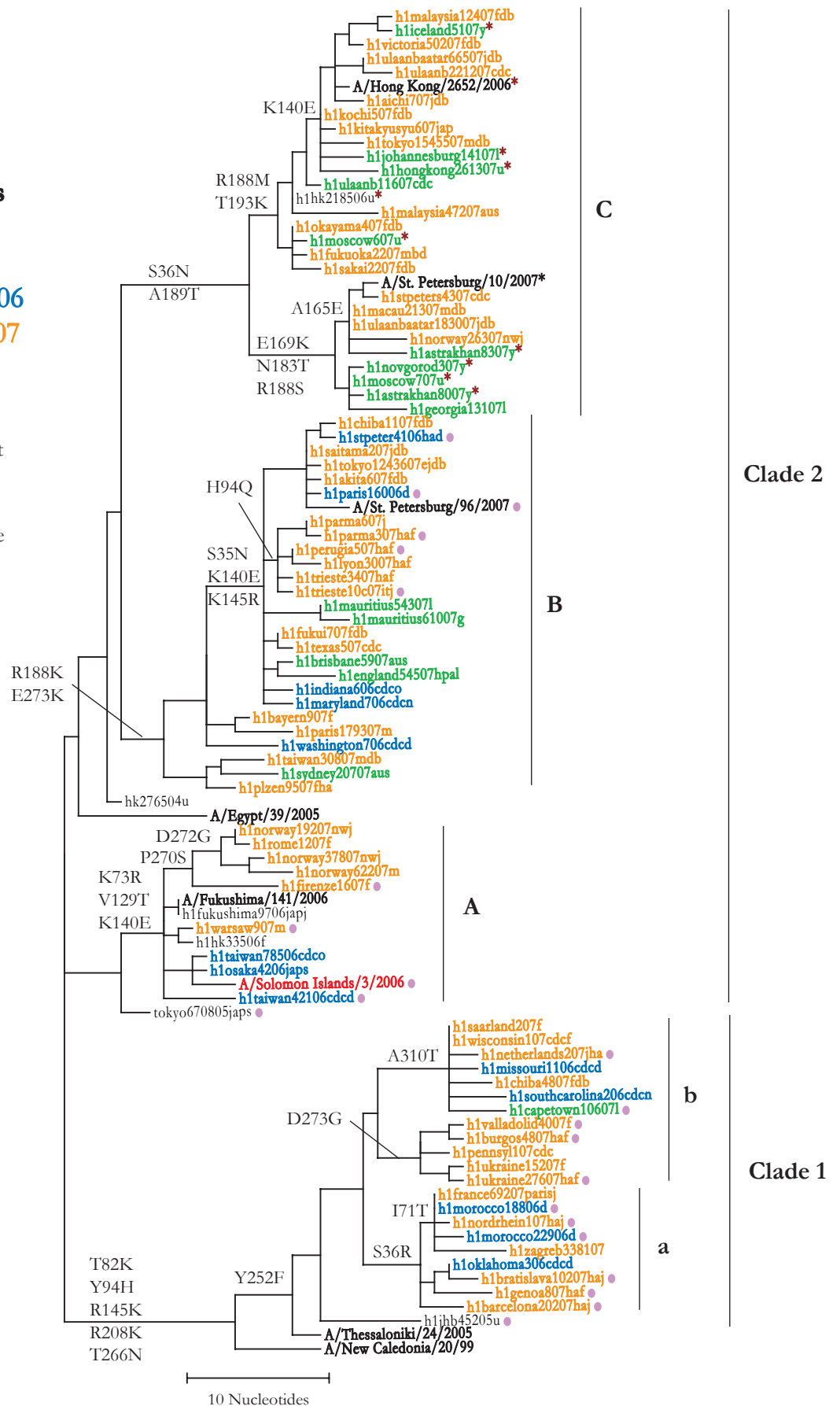


Figure 2. Phylogenetic comparison of N1 neuraminidases

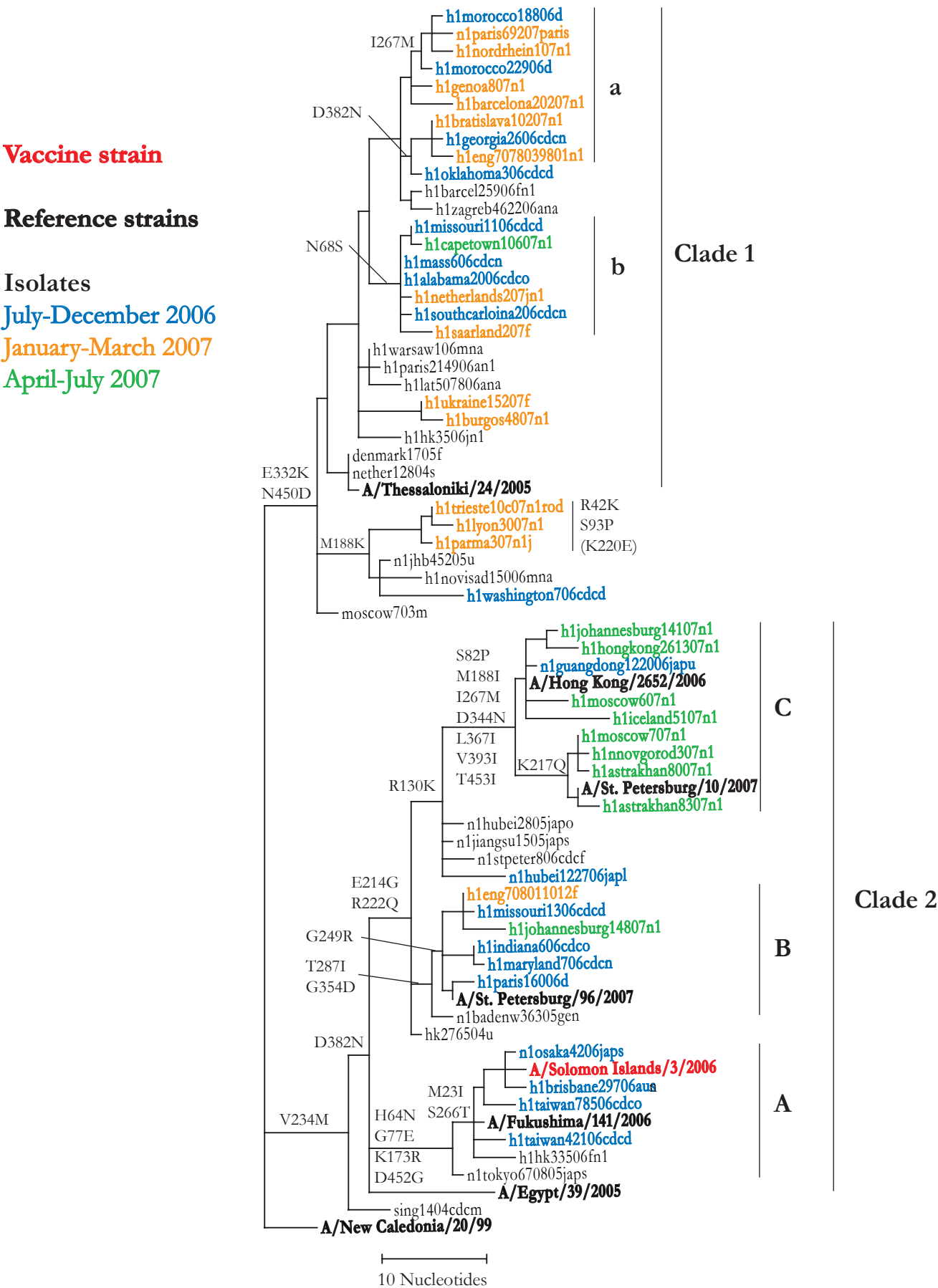


Table 4. Amino acid changes characteristic of the principal H1N1 sequence variants

Clade	Representative strain	Amino acid changes ¹	
		HA	NA
1 ²	A/Thessaloniki/24/05	Y252F	E332K N450D
		a (S36R) (I71T)	(I267M) (D382N)
		b (D273G) (A310T)	(N68S)
2	(Common changes)	T82K Y94H R145K R208K T266N	V234M D382N
2A	A/Solomon Islands/3/06	K73R V129T K140E (P270S) (+CHO) (D272G)	M23I H64N G77E K173R S266T D452G
2B	St.Petersburg/96/07	D35N K140E K145R R188K E273K	E214G R222Q G249R T287I G354D
2C	A/Hong Kong/2652/06	S36N R188M/S A189T (K140E) (A165E) (E169K) (N183T) (T193K)	S82P R130K M188I E214G (K217Q) R222Q I267M D344N L367I V393I T453I

1. Relative to A/New Caledonia/20/99, including common differences: HA, V165A and W251R; NA, V48I.

± CHO indicate addition or loss of a glycosylation site. 2. a and b refer to subgroups of clade 1.

Table 5. Antigenic analyses of influenza A H3N2 viruses (1)

Viruses	Isolation Date		Haemagglutination inhibition titre ¹									
			Post infecton ferret sera									
			A/Cal	A/HK	A/Hiro	A/Wis	A/Bay	A/Slov	A/Berlin	A/Lyon	A/Nepal	A/Sant
			7/04	4443/05	52/05	67/05	4/06	134/06	2/06	636/06	921/06	7981/06
			F31/05	F24/05	F2/06	F6/06	F11/06	F9/06	F8/06	F12/06	F2/07	Nib3/07
A/California/7/2004	16.9.04	SpfCk3, E3 \3	2560	320	320	640	1280	160	640	640	1280	640
A/Hong Kong/4443/2005	24.6.06	MDCK2 \6	320	640	640	320	320	320	40	80	320	640
A/Hiroshima/52/2005	25.11.06	E3 \3	640	640	2560	2560	640	320	80	320	1280	640
A/Wisconsin/67/2005	31.8.05	SpfCk3E3 \2	1280	1280	1280	2560	1280	320	80	320	1280	1280
A/Bayern/4/2006	7.3.06	MDCKx \5	640	160	80	160	1280	80	160	320	320	320
A/Slovakia/134/2006	6.2.06	MDCK1 \3	1280	640	160	640	640	160	40	320	1280	1280
A/Berlin/2/2006	6.2.06	MDCKx \3	320	160	80	160	320	80	80	160	320	320
A/Lyon/636/2006	15.2.06	MDCK2 \4	640	320	160	320	640	160	320	320	640	320
A/Nepal/921/2006	11.7.06	E2 \2	640	640	320	640	640	80	40	320	2560	640
A/Santiago/7981/2006	6.7.06	E5 \1	2560	640	320	1280	1280	160	40	640	2560	2560
A/Bratislava/46/07	9.1.07	E1 \1	640	640	320	1280	80	1280	10	40	160	160
A/Semily/55/2007	Feb-07	MDCK3, E2 \1	640	640	1280	1280	320	160	40	160	640	320
A/Prague/59/2007	Feb-07	MDCK4 \2	40	40	40	80	20	20	10	<	80	80
A/Algeria/453/06	Feb-07	MDCKx \2	40	160	40	80	20	20	<	<	80	80
A/Firenze/14/2007	Feb-07	MDCK2 \2	80	80	160	80	20	160	10	20	80	320
A/Vienna/338428/07	5.2.07	MDCK1 \2	320	320	320	320	80	320	10	160	640	640
A/Berlin/20/07	9.2.07	MDCKx \2	160	160	80	160	20	80	10	40	160	320
A/Lisbon/59/07	9.2.07	MDCKx \2	80	160	320	80	40	80	20	20	160	320
A/Zamora/35/07	9.2.07	MDCK1 \2	160	160	160	80	40	160	10	40	160	160
A/Eng/233/07	20.2.07	MDCKx \1	80	320	320	160	80	80	20	<	160	640
A/Valladolid/49/07	26.2.07	MDCK1 \2	640	640	320	640	160	640	40	320	640	640
A/Sachsen/8/07	6.3.07	MDCKx \2	160	640	320	80	80	320	20	40	640	320
A/Astrakhan/60/07	2.4.07	C1 \1	40	<	40	80	20	20	10	<	160	80
A/Ulan-Ude/1/07	2.4.07	C1 \2	320	640	640	1280	1280	160	160	320	1280	640
A/Voronezh/7/07	2.4.07	C1 \2	160	320	160	320	160	80	80	80	320	320
A/Tula/11/07	9.4.07	C1 \1	80	80	160	160	80	40	20	<	320	320
A/Khabarovsk/528/07	9.4.07	C1 \1	40	40	80	80	40	20	20	<	160	80

1. <, <40

Table 6. Antigenic analyses of influenza A H3N2 viruses (2)

Viruses	Isolation Date		Haemagglutination inhibition titre ¹									
			Post infecton ferret sera									
			A/Cal	A/HK	A/Wis	A/Bay	A/Berlin	A/Nepal	A/Sant	A/Parma	A/Trieste	A/Sofia
			7/04	4443/05	67/05	4/06	2/06	921/06	7981/06	1/07	25c/07	319/07
			F31/05	F24/05	F6/06	F11/06	F8/06	F2/07	Nib3/07	F7/07	F5/07	F6/07
A/California/7/2004	16.9.04	SpfCk3, E3 \3	2560	320	640	1280	640	640	640	160	320	160
A/Hong Kong/4443/2005	24.6.06	MDCK2 \6	160	320	320	320	20	320	640	160	160	160
A/Wisconsin/67/2005	31.8.05	SpfCk3E3 \2	1280	1280	2560	1280	80	1280	640	640	1280	1280
A/Bayern/4/2006	7.3.06	MDCKx \5	640	160	320	1280	320	640	640	160	160	160
A/Berlin/2/2006	6.2.06	MDCKx \3	320	160	80	640	160	320	640	80	80	160
A/Nepal/921/2006	11.7.06	E2 \2	1280	1280	1280	1280	40	5120	640	1280	1280	1280
A/Santiago/7981/2006	6.7.06	E5 \1	1280	640	1280	1280	20	1280	2560	320	1280	640
A/Parma/1/2007	Dec-06	MDCK1 \2	80	80	80	80	20	320	160	160	<	80
A/Trieste/25c/2007	Jan-07	MDCKx \2	<	40	<	40	10	40	160	<	40	40
A/Sofia/319/2007	Jan-07	E2 \2	1280	640	1280	1280	20	1280	1280	320	640	640
A/Paris/1716/2007	Feb-07	MDCKx \1	80	160	160	80	20	160	160	80	80	80
A/Sofia/671/2007	Feb-07	MDCK1, E1 \3	640	320	80	320	320	40	<	160	1280	640
A/Zagreb/3064/07	Feb-07	E	640	320	320	640	10	1280	640	320	640	640
A/Norway/414/2007	12.2.07	MDCK1 \1	320	160	160	640	80	160	320	80	80	160
A/Kiev/546/07	12.3.07	MDCKx \1	640	1280	640	640	—	160	640	1280	1280	640
A/Borisov/175/07	13.3.07	MDCK1 \1	40	40	40	20	10	40	80	<	<	<
A/Dniprodzerzinsk SES/350/07	16.3.07	MDCKx \1	40	80	80	40	—	10	80	40	40	80
A/Belarus/340/2007	23.3.07	MDCK1 \1	40	80	80	20	10	80	160	40	40	40
A/Latvia/4897/07	2.4.07	MDCK1 \1	80	160	160	80	20	160	160	80	40	160
A/Ulan-Ude/1/07	2.4.07	C1 \2	320	640	640	1280	1280	160	160	320	1280	640
A/Norway/787/2007	4.4.07	-MK2, MDCK1 \1	40	80	80	40	10	80	0	40	80	40
A/Mauritius/307/07	24.4.07	MDCK1 \1	40	80	80	40	20	80	160	<	40	40
A/Mauritius/354/2007	15.5.07	MDCK2 \1	40	160	160	160	—	320	320	—	320	160
A/St Petersburg/95/2007	21.5.07	C3 \1	40	80	40	40	—	160	80	—	160	80
A/Moscow/15/2007	21.5.07	C3 \2	<	40	40	20	—	80	80	—	40	<
A/Hong Kong/2571/2007	28.5.07	MDCK2 \1	<	40	20	20	—	40	40	—	<	40
A/Hong Kong/2657/2007	8.6.07	MDCK2 \1	<	40	20	10	—	80	80	—	<	40

1. <, <40

Table 7. Antigenic analyses of influenza A H3N2 viruses (3)

Viruses	Isolation Date		Haemagglutination inhibition titre ¹									
			Post infection ferret sera									
			A/Wis	A/Nepal	A/Sant	A/Parma	A/Trieste	A/Sofia	A/Zag	A/Serr	A/Prag	A/Tri
			67/05	921/06	7981/06	1/07	25c/07	319/07	1216/07	77/07	3/07	25E/07
			F6/06	F2/07	Nib3/07	F7/07	F5/07	F6/07	F12/07	F10/07	F9/07	F11/07
A/Wisconsin/67/2005	31.8.05	SpfCk3E3 \3	2560	5120	2560	1280	2560	1280	2560	1280	640	2560
A/Nepal/921/2006	11.7.06	E2 \2	1280	5120	2560	1280	2560	1280	1280	1280	1280	2560
A/Santiago/7981/2006	6.7.06	E5 \1	1280	2560	2560	640	2560	1280	1280	320	640	1280
A/Parma/1/2007	Jan-07	MDCK1 \2	160	640	320	320	80	160	40	160	80	80
A/Trieste/25c/2007	Jan-07	MDCKx \2	40	80	160	40	80	40	<	80	<	80
A/Sofia/319/2007	Jan-07	E2 \2	1280	1280	2560	640	1280	1280	1280	320	640	1280
A/Zagreb/1216/2007	Jan-07	Ex \1	1280	1280	1280	320	1280	640	1280	320	640	1280
A/Serres/77/2007	25.1.07	E1 \1	2560	2560	1280	640	1280	1280	1280	2560	640	1280
A/Prague/3/2007	Feb-07	MDCK3, E4 \2	1280	2560	1280	640	1280	640	640	640	640	1280
A/Trieste/25E/2007	Jan-07	MDCKx \2,E1	1280	2560	1280	640	1280	640	1280	320	640	1280
A/Ulaanbaatar/1106/2007	unknown	E3 \1	80	160	160	80	40	80	40	80	40	80
A/Santiago/6881/2007	unknown	C3 \1	80	160	160	80	80	160	40	80	80	160
A/Henan/Jinshui/147/2007	unknown	E3 \1	640	1280	1280	320	1280	640	640	320	320	640
A/Wisconsin/3/2007	21.1.07	Ex \2	2560	5120	2560	1280	5120	2560	2560	1280	640	2560
A/Iceland/19/07	12.2.07	MDCK1 \1	640	5120	1280	640	1280	1280	640	640	640	1280
A/Georgia/13/2007	21.2.07	MDCK2 \1	320	640	640	320	640	320	160	320	160	640
A/Georgia/55/2007	2.3.07	MDCK3 \1	80	80	80	<	80	40	<	80	<	80
A/Rzeszow/2/07	2.3.07	MDCK \1	80	80	160	40	80	80	<	160	<	80
A/Iceland/46/07	17.3.07	MDCK2 \1	40	160	80	80	40	40	<	80	<	80
A/Georgia/135/2007	4.4.07	MDCK2 \1	640	5120	1280	1280	2560	2560	2560	2560	1280	2560
A/Iceland/48/07	5.4.07	MDCK2 \1	40	80	80	40	80	40	<	40	<	80
A/Sweden/4/07	16.4.07	MDCK2 \2	160	640	160	160	320	160	80	80	80	320
A/Sweden/2/07	16.4.07	MDCK2 \1	160	320	320	160	80	160	<	160	80	80
A/Stockholm/6/07	16.4.07	MDCK1 \2	320	320	320	160	80	160	320	160	80	320
A/Stockholm/7/07	18.4.07	MDCK1 \1	80	160	160	80	80	160	<	80	<	80
A/Stockholm/8/07	18.4.07	MDCK1 \2	40	80	160	40	40	40	<	40	<	80
A/Cape Town/97/07	July-07	MDCKx \1	80	320	320	160	80	160	80	160	40	160
A/Johannesburg/46/07	1.7.07	MDCKx \1	80	160	160	80	80	80	<	80	<	80
A/Johannesburg/74/07	9.7.07	MDCKx \1	80	160	320	80	80	160	<	160	<	160

1. <, <40

Table 8. Antigenic analyses of influenza A H3N2 viruses (4)

Viruses	Isolation Date		Haemagglutination inhibition titre ¹									
			Post infection ferret sera									
			A/Wis 67/05	A/Nepal 921/06	A/Parma 1/07	A/Trieste 25c/07	A/Sofia 319/07	A/Serr 77/07	A/Prag 3/07	A/Tri 25E/07	A/Wis 3/07	A/Hen 147/07
			F6/06	F2/07	F7/07	F5/07	F6/07	F10/07	F9/07	F11/07	F15/07	F16/07
A/Wisconsin/67/2005	31.8.05	SpfCk3E3 \3	5120	5120	1280	2560	2560	1280	640	2560	1280	5120
A/Nepal/921/2006	11.7.06	E2 \2	2560	5120	2560	2560	2560	1280	1280	2560	1280	5120
A/Parma/1/2007	Jan-07	MDCK1 \2	320	640	320	320	160	160	160	320	160	640
A/Trieste/25c/2007	Jan-07	MDCKx \2	80	320	80	160	80	80	<	160	160	160
A/Sofia/319/2007	Jan-07	E2 \2	2560	5120	640	2560	1280	640	640	1280	640	5120
A/Serres/77/2007	25.1.07	E1 \1	2560	5120	640	2560	1280	640	640	1280	640	5120
A/Prague/3/2007	Feb-07	MDCK3, E4 \2	2560	5120	1280	2560	2560	640	1280	2560	640	5120
A/Trieste/25E/2007	Jan-07	MDCKx \2,E1	2560	5120	640	2560	2560	640	640	2560	640	5120
A/Wisconsin/3/2007	21.1.07	Ex \2	5120	5120	1280	2560	2560	1280	1280	2560	1280	5120
A/Henan/Jinshui/147/2007	unknown	E3 \1	1280	5120	640	1280	640	320	640	1280	320	5120
A/Georgia/25/2007	23.2.07	MDCK2 \2	160	320	80	80	80	80	40	80	160	160
A/Georgia/59/2007	6.3.07	MDCK2 \2	160	320	160	80	80	80	80	80	160	160
A/Perth/27/2007	5.7.07	MDCKx+2 \1	160	640	80	160	80	80	40	160	160	160
A/Brisbane/48/2007	26.6.07	MDCKx+1 \1	<	160	40	40	40	40	<	80	80	80
A/Philippines/1616/2007	28.5.07	MDCK4 \1	80	320	80	40	80	80	40	80	80	80
A/Sydney/230/2007	4.7.07	MDCK2 \1	40	160	40	80	80	80	<	160	160	80
A/Brisbane/10/2007	6.2.07	E2 \1	1280	5120	320	1280	640	320	320	1280	320	2560
A/New Caledonia/9/2006	24.10.06	E7 \1	2560	5120	640	2560	1280	640	320	1280	640	2560
A/Brisbane/4/2007	18.1.07	E2 \1	2560	5120	1280	2560	1280	640	640	2560	1280	5120
A/Ulaanbaatar/1106/2007	unknown	E3 \1	160	320	160	320	160	320	80	160	160	320
A/Santiago/6881/2007	unknown	C3 \1	160	160	80	160	80	160	40	160	160	160
A/Kobe/9/2007	26.1.07	MDCK3 \1	40	80	<	<	80	40	<	<	160	80
A/Osaka/29/2007	21.2.07	MDCK4 \1	80	80	40	80	40	80	<	160	160	80
A/Johannesburg/91/2007	Jul-07	MDCK \1	40	80	40	80	80	80	<	80	160	160
A/Hong Kong/1957/2007	28.3.07	MDCK2 \1	80	80	40	160	40	80	<	160	80	320
A/Borisov/158/2007	13.3.07	MDCK1 \1	80	80	40	40	80	80	40	40	160	160
A/Stockholm/3/2007	16.04.07	MDCK0 \2	160	160	80	80	80	80	40	160	80	80
A/Johannesburg/66/2007	3.7.07	MDCK \1	160	160	80	80	80	80	40	160	160	80
A/Johannesburg/92/2007	Jul-'07	MDCK \1	160	160	80	80	80	80	40	160	160	160
A/Iceland/48/2007	5.4.07	MDCK2 \1	160	160	80	160	80	160	40	160	160	160
A/Georgia/125/2007	26.03.2007	MDCK2 \1	160	160	80	160	80	160	40	160	160	160

Figure 3. Phylogenetic comparison of H3 HAs

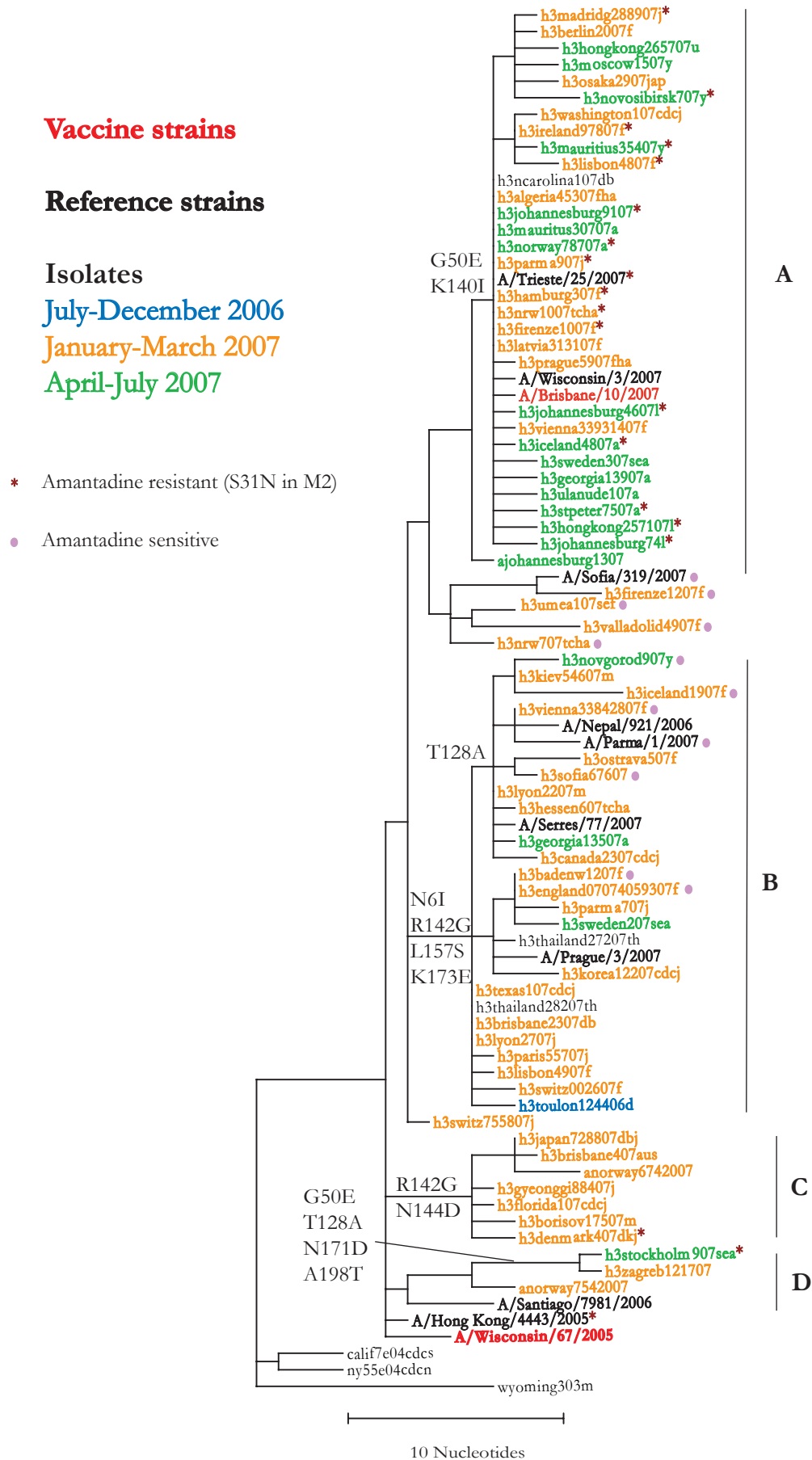


Figure 4. Phylogenetic comparison of N2 neuraminidase genes

Vaccine strains

Reference strains

Isolates

July-December 2006

January-March 2007

April-July 2007

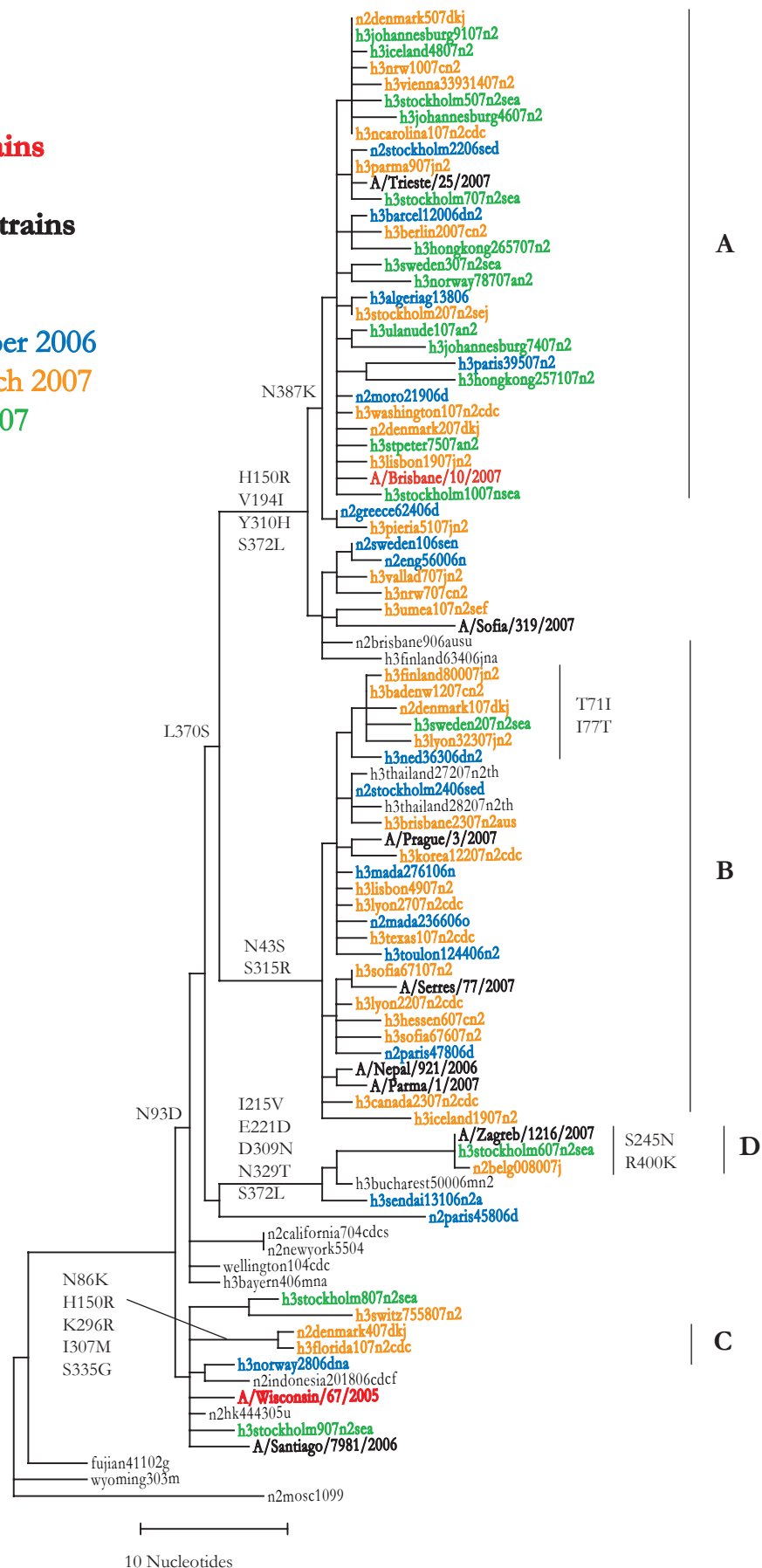


Table 9. Amino acid changes characteristic of the principal H3N2 sequence variants

Clade	Representative strain	Amino Acid changes ¹	
		HA	NA
A	A/Trieste/25/07	G50E K140I (R261Q)	H150R V194I Y310H L370S S372L N387K
B	A/Nepal/921/06	N6I R142G L157S K173E (T128A)(-CHO)	N43S S315R L370S (T71I) (I77T)
C	A/Denmark/4/07	R142G N144D(-CHO)	N86K(-CHO) H150R K296R I307M S335G
D	A/Zagreb/1216/07	G50E T128A(-CHO) N171D A198T	I215V E221D S245N (+CHO) D309N(+CHO) N329T(-CHO) S372L R400K

1. Relative to A/Wisconsin/67/05. ± CHO indicates addition or loss of a glycosylation site.

Table 10. Antigenic analyses of influenza B viruses (Victoria lineage)

Viruses	Isolation date	Passage History	Haemagglutination inhibition titre ¹					
			B/Shan ² 7/97	Post infection ferret sera				
				B/Shan 7/97	B/Bris ³ 32/02	B/HK 45/05	B/Mal 2506/04	B/Vic 304/06
B/Shandong/7/97		Ex	2560	320	160	160	320	320
B/Brisbane/32/2002	2.7.02	E2 \6	1280	320	160	160	320	160
B/HK/45/2005	7.2.05	MDCK2 \4	2560	320	160	160	320	320
B/Malaysia/2506/2004	6.12.04	E3 \3	1280	320	160	160	320	320
B/Victoria/304/2006	6.6.06	E2 \2	640	160	80	80	160	160
B/Pennsylvania/5/2007	unknown	E3 \1	1280	320	80	80	160	160
B/Roma-ISS/2/2007	Feb-07	MDCK1 \2	1280	320	40	160	80	160
B/Astrakhan/2/2007	5.2.07	C2 \1	1280	320	40	160	160	160
B/St. Petersburg/17/2007	9.2.07	C2 \1	2560	320	80	320	80	320
B/Odessa/Ukr/149/2007	14.2.07	MDCKx \1	640	160	80	80	160	160
B/St.Petersburg/15/2007	19.2.07	C2 \1	1280	320	20	160	40	40
A/Georgia/12/2007	21.2.2007	MDCK3 \1	1280	320	40	80	160	80
B/Dnipropetrovsk/335/2007	28.2.07	MDCKx \1	1280	160	80	80	160	320
B/Albania/153/2007	6.3.07	MDCK2 \1	1280	320	<	160	40	40
B/Hong Kong/121/2007	7.3.07	MDCK2 \1	1280	320	20	160	320	160
B/Lisbon/2/2007	9.3.07	MDCK1 \1	2560	320	40	160	40	80
B/Irkutsk/97/2007	12.3.07	C1 \1	1280	160	10	160	40	40
B/Osaka/16/2007	13.3.07	MDCK4 \1	1280	320	80	80	320	160
B/Odessa/394/2007	19.3.07	MDCKx \1	2560	320	80	160	40	80
B/Madagascar/393/2007	22.3.07	MDCK1 \1	1280	320	10	160	40	40
B/Kaliningrad/11/2007	2.4.07	C1 \3	2560	320	80	320	80	160
B/Khabarovsk/26/2007	2.4.07	C2 \1	1280	320	10	160	40	40
A/Voronezh/8/2007	2.4.07	C1 \1	1280	160	<	160	40	40
B/Tula/3/2007	9.4.07	C1 \3	2560	320	<	320	40	80
A/Tver/511/2007	9.4.07	C1 \1	1280	320	<	160	40	40
B/Astrakhan/88/2007	4.6.07	C3 \1	2560	320	160	640	80	320
B/Mauritius/468/2007	4.6.07	MDCK1 \1	1280	320	80	160	320	160
B/Dakar/5/2007	5.6.07	MDCK2 \2	1280	320	10	160	<	40
B/Johannesburg/18/2007	7.6.07	MDCKx \1	1280	320	20	160	40	80
B/Mauritius/498/2007	13.7.07	MDCK2 \1	1280	640	10	160	40	40
B/Victoria/500/2007	17.7.07	RMIX-x, MDCK1 \1	1280	80	<	160	40	40
B/Johannesburg/150/2007	23.7.07	MDCK \1	1280	80	<	160	40	40

¹ <, <40

² hyperimmune sheep serum

³ <, <10

Figure 5. Phylogenetic comparison of B HAs (Victoria lineage)

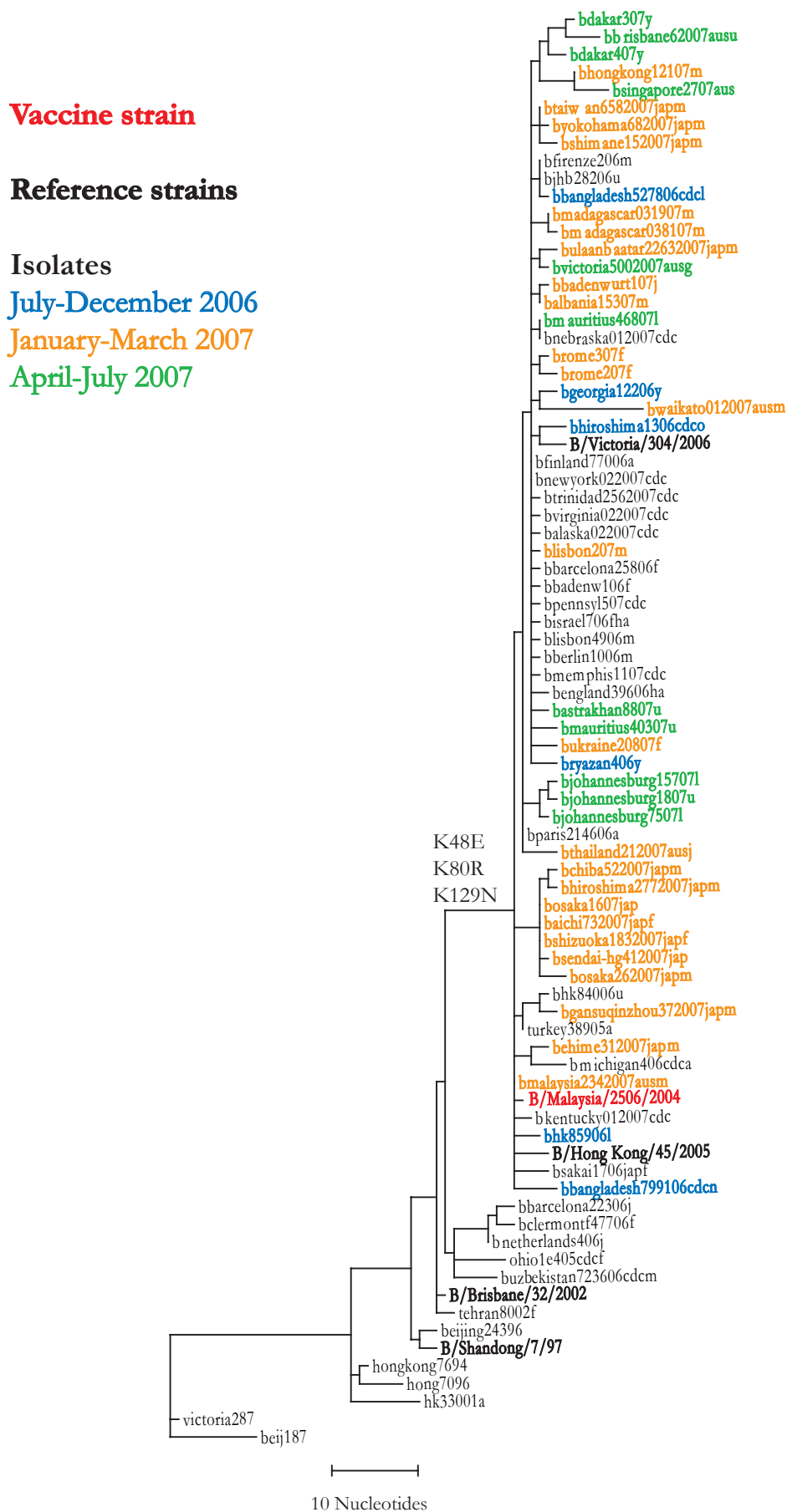


Table 11. Antigenic analyses of influenza B viruses (Yamagata lineage)

Viruses	Collection date	Passage History	Haemagglutination inhibition titre ¹					
			Post infection ferret sera					
			B/Sich 379/99	B/Shai 361/02	B/Jiang 10/03	B/Eg 144/05	B/FI 7/04	B/FI 4/06 CDC
B/Sichuan/379/99		Ex	320	160	40	320	320	160
B/Shanghai/361/2002	12.6.02	E3 \4	640	320	80	640	640	320
B/Jiangsu/10/2003	12.9.03	E4 \3	20	<	320	80	80	320
B/Egypt/144/2005	5.1.05	E2 \2	320	160	40	640	640	160
B/Florida/7/2004	1.10.05	E3 \2	320	160	80	640	640	320
B/Florida/4/2006		E2 \1	160	80	40	320	640	160
B/Chongqing/Yongchuan/18/2007	unknown	E3 \1	<	80	10	80	160	160
B/Soligorsk/140/2007	unknown	C2 \3	320	640	320	320	1280	640
B/Chungbuk/2274/2007	25.01.07	MDCK2 \1	80	80	320	640	640	—
B/Shizuoka/109/2007	26.1.07	MDCK3 \1	80	160	320	160	1280	1280
B/Roma-ISS/5/2007	Feb-07	MDCK1 \2	160	80	40	320	320	—
B/Novosibirsk/2/2007	9.2.07	C2 \1	80	80	40	160	160	—
B/Jeju/2787/2007	27.2.07	MDCK2 \1	40	80	80	320	320	—
B/Lisbon/1/2007	13.2.07	MDCK1 \1	40	80	80	160	160	—
B/Albania/130/2007	1.3.07	MDCK2 \1	80	160	80	160	320	—
B/Albania/134/2007	2.3.07	MDCK2 \1	80	160	80	160	320	—
B/Norway/661/2007	8.3.07	MDCK1 \1	160	160	320	320	320	—
B/Brisbane/3/2007	9.3.07	E2 \1	80	320	80	160	640	160
B/Belarus/338/2007	23.3.07	MDCK1 \1	320	640	160	320	640	—
B/Norway/794/2007	26.3.07	MDCK1 \1	40	80	80	160	160	—
B/Stockholm/1/2007	16.4.07	MDCK1 \1	40	40	80	160	160	—
B/Hong Kong/312/2007	17.4.07	MDCK2 \1	160	320	320	640	640	—
B/Sendai-H/114/2007	30.4.07	E3 \1	40	80	20	160	160	160
B/Hong Kong/441/2007	10.5.07	MDCK2 \1	160	160	320	640	640	—
B/Chelyabinsk/306/2007	4.6.07	C2 \1	320	160	320	640	640	—
B/Perth/12/2007	18.6.07	MDCKx+1 \1	40	80	320	160	640	640
B/Mauritius/481/2007	9.7.07	MDCK1 \1	40	80	80	160	640	1280
B/Johannesburg/90/2007	12.7.07	MDCK \1	40	80	80	80	160	—
B/Johannesburg/1197/2007	12.7.07	MDCKx+1 \1	80	80	160	160	640	320

¹ <, <40

Figure 6. Phylogenetic comparison of B HAs (Yamagata lineage)

Vaccine strain

Reference strains

Isolates

July-December 2006

January-March 2007

April-July 2007

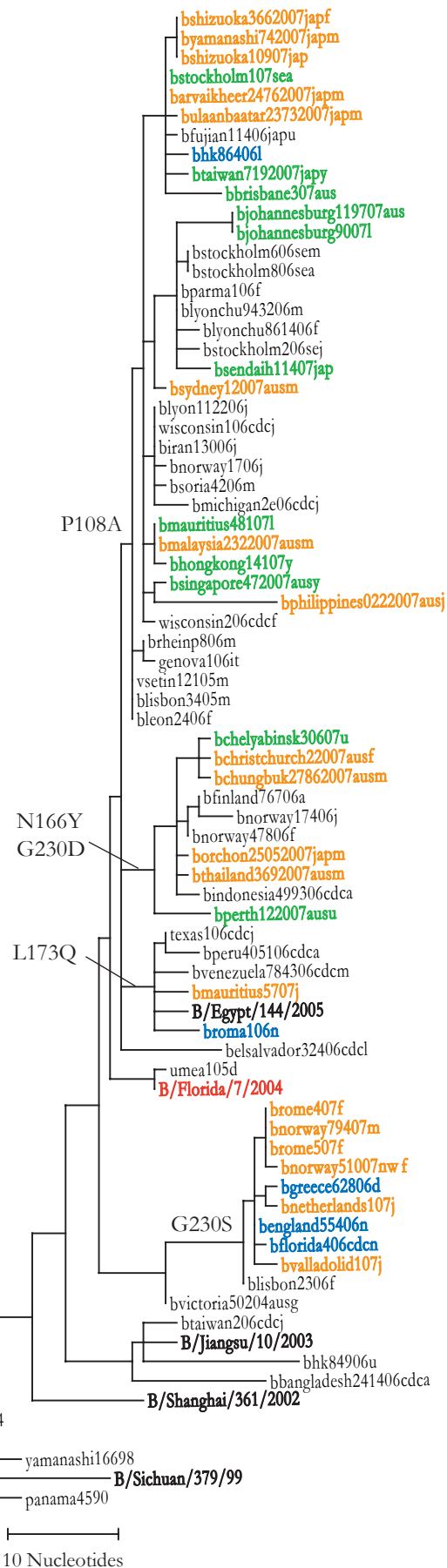


Figure 7. Phylogenetic comparison of B neuraminidase genes

Vaccine strains

Reference strains

Isolates

July-December 2006

January-March 2007

April-July 2007

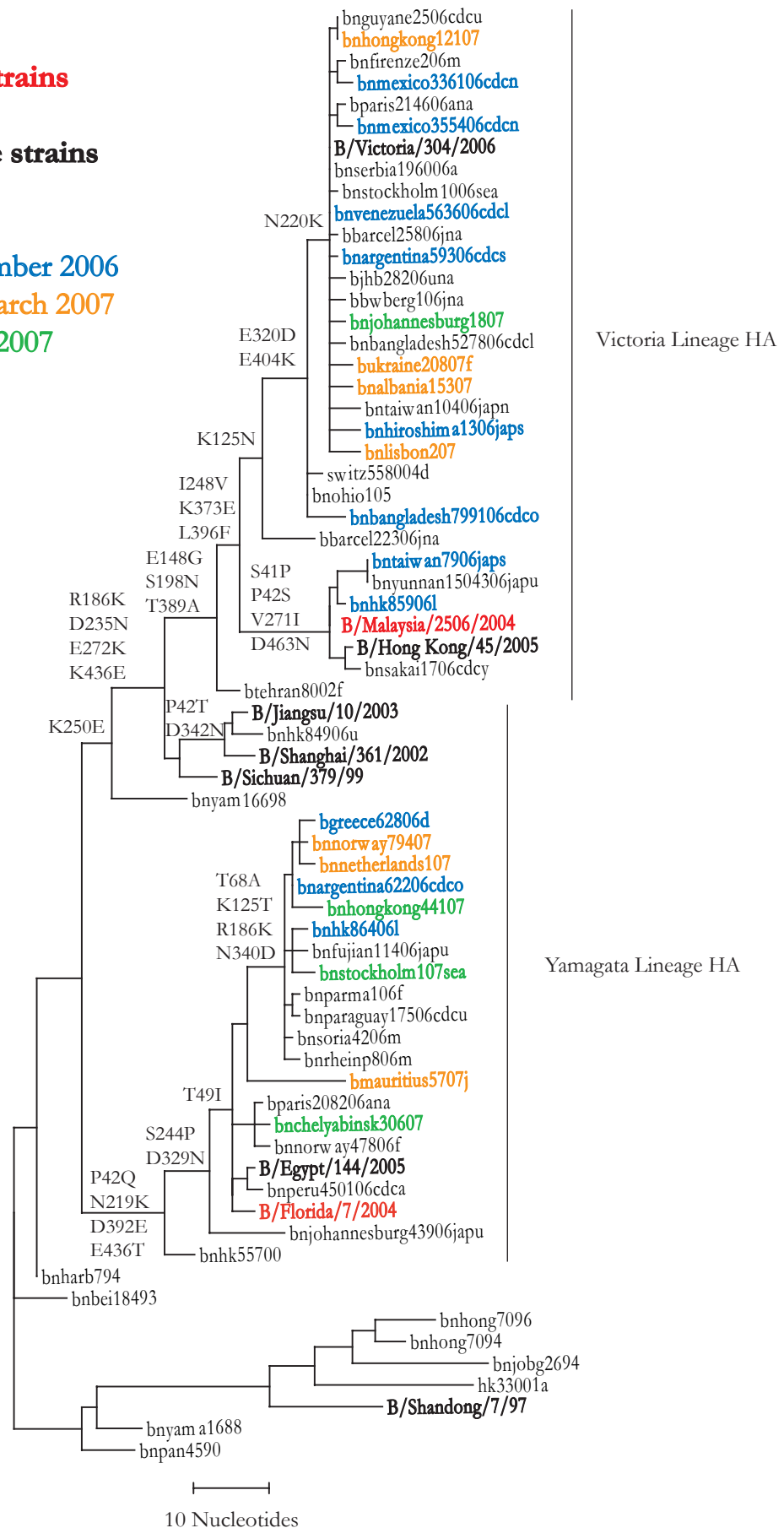


Table 12. Amino acid changes characteristic of recent B sequence variants

HA Lineage	Representative strain	Amino acid changes	
		HA	NA
Yamagata ¹	B/Chelyabinsk/306/07	N166Y (-CHO) G230D	—
	B/Hong Kong/864/06	P108A	T68A K125T R186K N340D
	B/Netherlands/1/07	G230S	T68A K125T R186K N340D
Victoria ²	B/Victoria/304/06	—	P41S S42P K125N N220K I271V E320D E404K N463D

1. Relative to B/Florida/7/04. 2. Relative to B/Malaysia/2506/04.